THÉO CASTILLO

AI/ML Engineering Apprentice at the French Space Agency (CNES)
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WORK EXPERIENCE

AI/ML Engineering Apprentice, Brain-Computer Interfaces — Sept. 2026 – 2029
CNES (French Space Agency), Toulouse, France

- Development of a brain signal processing and classification prototype using ML/DL algorithms, running on a Linux HPC cluster
- Business needs analysis and identification of relevant use cases for AI integration in the space domain (robotic tool control, cognitive monitoring of operators and astronauts)
- State of the art review of brain-computer interface technologies applied to the space sector

AI Developer, Work-Study Program — Sept. 2025 – June 2026
Genial (AI startup), Bordeaux, France

- Design and deployment of specialized AI agents for B2B clients, from requirement analysis to production
- Architecture and management of RAG pipelines with context engineering, multi-agent orchestration and MCP servers
- Containerization with Docker, CI/CD pipeline setup for robust deployment
- Iterative client feedback management and continuous adaptation of deployed solutions

AI Agent Developer, Freelance — June – July 2025
Genial, Bordeaux, France

- Creation and maintenance of specialized AI agents with full autonomy
- Setup and management of a sole proprietorship (first entrepreneurial experience)

Machine Learning Research Intern — Apr. – June 2025
Bekki Lab, Ochanomizu University, Tokyo, Japan

- Reproduced the GPT-2 architecture in Haskell using Hasktorch (functional programming approach)
- Worked in an NLP/ML research environment, exploring formal approaches to deep learning

EDUCATION

Engineering Degree, Digital Systems (Work-Study) — 2026 – 2029
INP-ENSEEIHT (N7), Toulouse

Bachelor of Computer Science, International Track — 2023 – 2026
IUT de Gradignan, University of Bordeaux
Ranked 1st out of 66 students (Semester 5).

Technology Baccalaureate STI2D, with Highest Honors — 2020 – 2023
Lycée Gustave Eiffel, Bordeaux

TECHNICAL PROJECTS (tcastillo.me)

VeridisQuo, deepfake detection [Python · PyTorch · YOLO · OpenCV · FastAPI]
Application combining spatial and frequency analysis. Face detection with YOLO, PyTorch classifier, real-time visualization and FastAPI endpoint. 60 stars on GitHub.

Neural Networks Library from scratch [Python · NumPy]
NumPy implementation: linear layers, activation functions, loss functions, optimizers and backpropagation.

Deep Rubiks [Python · PyTorch · React · Three.js · FastAPI]
Implementation of the DeepCubeA algorithm to solve the Rubik's Cube using a neural network and A* search, with 3D visualization in Three.js.

TECHNICAL SKILLS

Languages: Python, Haskell, C, C++, TypeScript, JavaScript, Java, Bash
AI / ML: PyTorch, scikit-learn, LangChain, NumPy, Pandas, Matplotlib, OpenCV, YOLO, Gymnasium, TensorFlow
Tools / DevOps: Docker, Git, GitHub/GitLab, Terraform, AWS, Scaleway, MCP, Jupyter, n8n, LangSmith
Databases: PostgreSQL, MySQL, SQLite
Web / API: FastAPI, React, Flask, Node.js, Three.js

LANGUAGES & INTERESTS

Languages: French (native), English B2, Spanish A2
Interests: Formula 1, chess, running, collectible card games, myrmecology
Community: Kaggle competitions, published a Formula 1 dataset on Kaggle